Coding & Solution

Date: 15 March 2026

Team ID: XXXXX

Project Name: Smart Lender – Loan Eligibility Prediction System

Maximum Marks: 5 Marks

Coding & Solution

The **Smart Lender – Loan Eligibility Prediction System** was developed as an **individual project** using Python, Flask, HTML, CSS, and Machine Learning. The application automates the loan eligibility prediction process by accepting applicant details, preprocessing the data, and using a trained **Random Forest Classifier** to predict whether a loan application is likely to be approved or rejected. The solution is modular, easy to maintain, and follows good coding practices to ensure reliability and readability.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/bhargavi-glitch/Smart-Lender
Render Link/ URL	https://smart-lender-16bp.onrender.com/
Programming Language(s)	Python 3.x, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Machine Learning Algorithm	Random Forest Classifier
Key Features Implemented	Home Page, Loan Application Form, Input Validation, Data Preprocessing, Loan Eligibility Prediction, Result Display, Flask Backend Integration
Pending / Future Features	User Authentication, Database Integration, Cloud Deployment, Mobile Application, Real-Time Credit Score Integration
Setup / Run Instructions	Install the required Python libraries, place the trained model.pkl model in the project directory, run python app.py, and open http://127.0.0.1:5000 in a web browser.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Comments and documentation are included where required	Yes
4	Input validation and error handling are implemented	Yes
5	The application runs without critical errors	Yes

Technologies Used

- **Programming Language:** Python
- **Frontend:** HTML5, CSS3
- **Backend:** Flask
- **Machine Learning:** Random Forest Classifier
- **Libraries:** Pandas, NumPy, Scikit-learn, Pickle
- **Development Environment:** Visual Studio Code
- **Version Control:** Git & GitHub (if applicable)

Solution Workflow

1. Launch the Smart Lender application.
2. Open the loan application form.
3. Enter the applicant's personal and financial details.
4. Submit the form for validation.
5. The Flask backend preprocesses the input data.
6. The trained **Random Forest (model.pkl)** model predicts the loan eligibility.
7. The system displays **Loan Approved** or **Loan Rejected** on the result page.